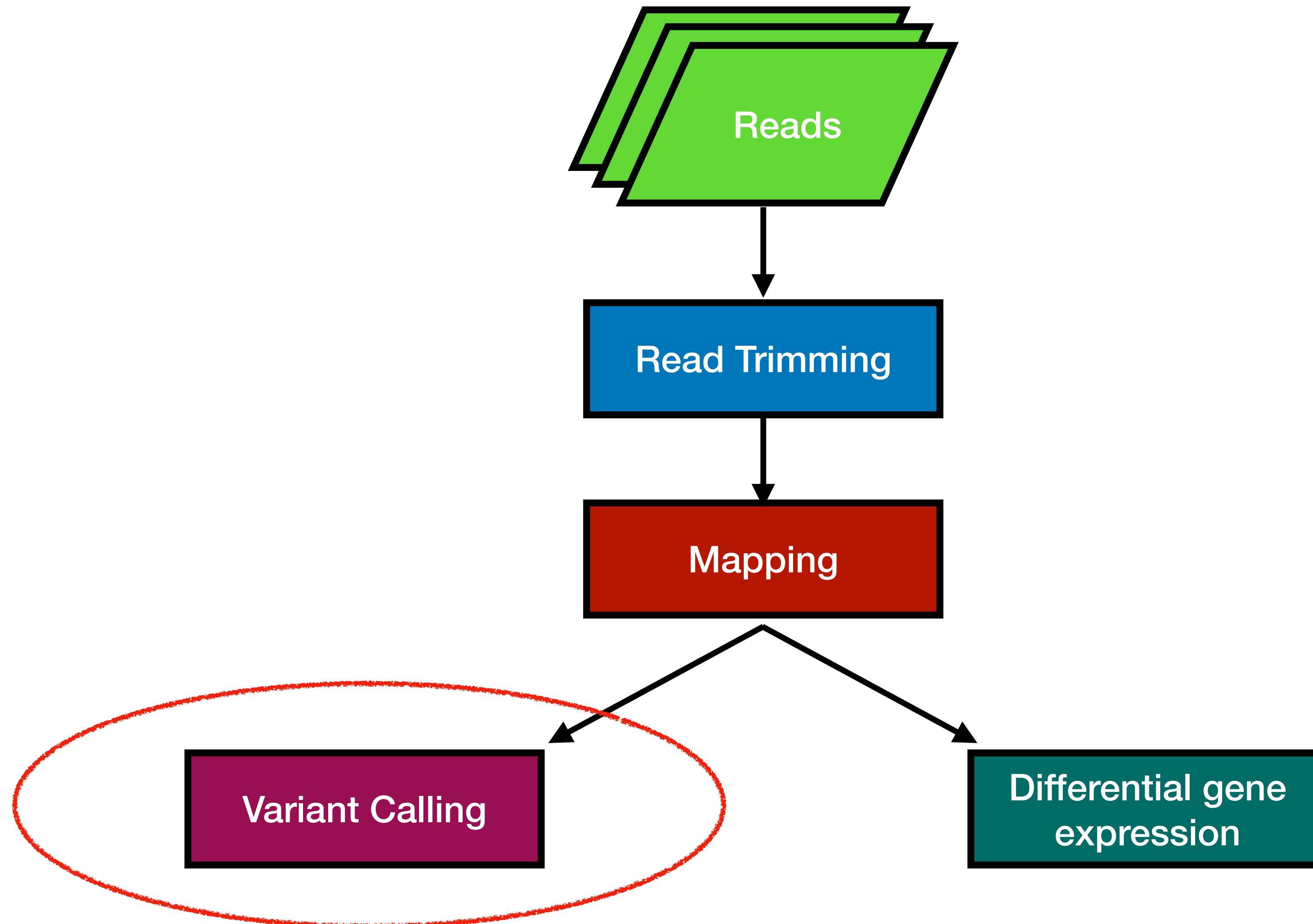


Variant Calling and Annotation

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NGS Workflows



Agenda for today

- Terminology
- Variant calling file format
- Workflows
- Applications
- Variant calling in diagnostics

Variant Calling

Detection of nucleotide differences between the reads of a sample and a reference sequence

- Single Nucleotide Polymorphism (SNP)

AACGTAGCTAG
AACGTGGCTAG

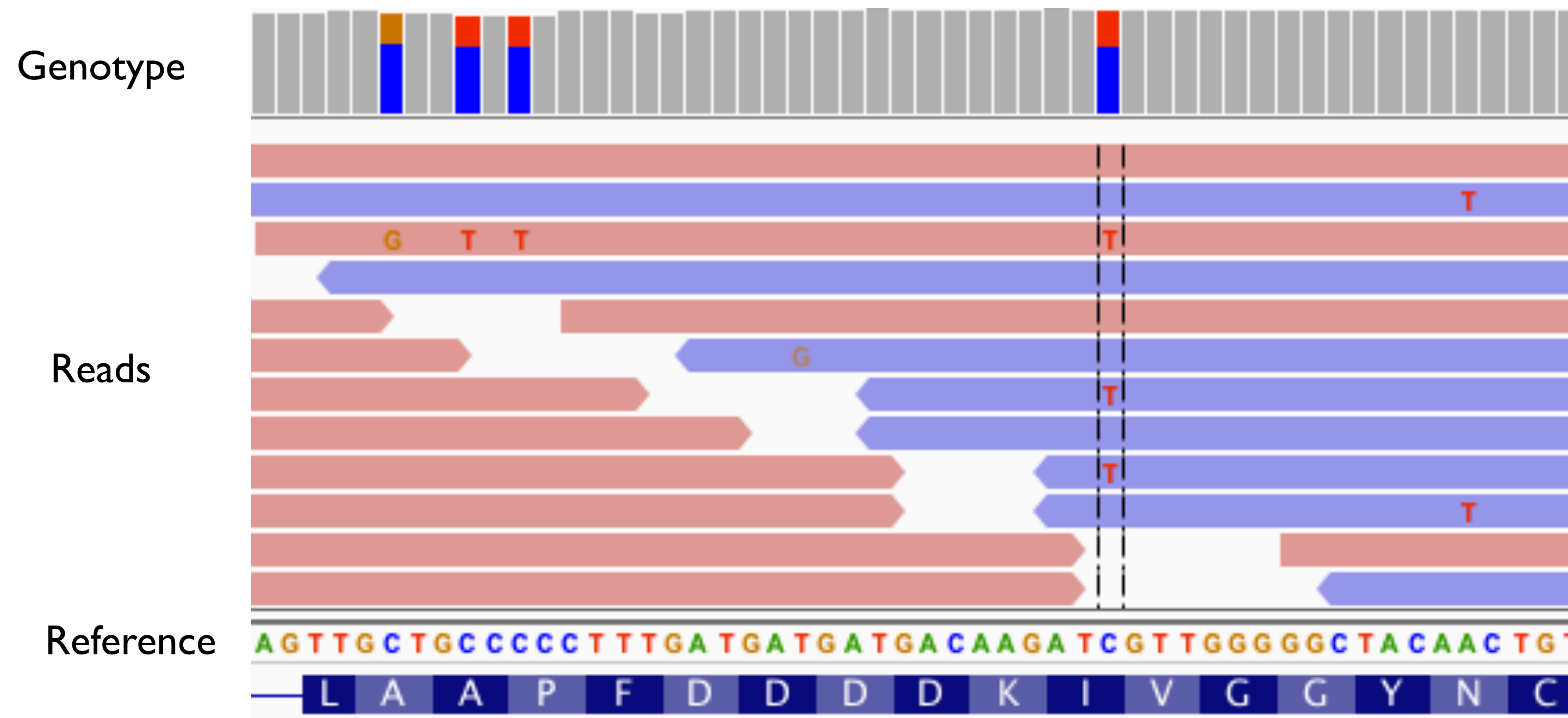
- Insertion/Deletion (InDel)

AACGTAGGCTAG
AACGT — GCTAG

Larger variations -> Structural Variant analysis

Variant Calling

Detection of SNPs & InDels and Genotype estimation



Variant Calling Format

- Full specifications: <http://samtools.github.io/hts-specs/>
- Text file format tabs-separated:
 - meta-information lines (start with “##”)
 - a header line (starts with “#”)
 - a separate line for each variant detected
- Contains **information about one or more samples**

Variant Calling Format

Metadata

```
##fileformat=VCFv4.4
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
```

Header

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	NA00001	NA00002	NA00003
20	14370	rs6054257	G	A	29	PASS	NS=3;DP=14;AF=0.5;DB;H2	GT:GQ:DP:HQ	0 0:48:1:51,51	1 0:48:8:51,51	1/1:43:5:.,.
20	17330	.	T	A	3	q10	NS=3;DP=11;AF=0.017	GT:GQ:DP:HQ	0 0:49:3:58,50	0 1:3:5:65,3	0/0:41:3
20	1110696	rs6040355	A	G,T	67	PASS	NS=2;DP=10;AF=0.333,0.667;AA=T;DB	GT:GQ:DP:HQ	1 2:21:6:23,27	2 1:2:0:18,2	2/2:35:4
20	1230237	.	T	.	47	PASS	NS=3;DP=13;AA=T	GT:GQ:DP:HQ	0 0:54:7:56,60	0 0:48:4:51,51	0/0:61:2
20	1234567	microsat1	GTC	G,GTCT	50	PASS	NS=3;DP=9;AA=G	GT:GQ:DP	0/1:35:4	0/2:17:2	1/1:40:3

Entries

Variant Calling Format

INFO

```
##fileformat=VCFv4.4
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##source=myImputationProgramV3.1
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#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT NA00001 NA00002 NA00003
20 14370 rs6054257 G A 29 PASS NS=3;DP=14;AF=0.5;DB;H2 GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20 17330 . T A 3 q10 NS=3;DP=11;AF=0.017 GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3 0/0:41:3
20 1110696 rs6040355 A G,T 67 PASS NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2 2/2:35:4
20 1230237 . T . 47 PASS NS=3;DP=13;AA=T GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20 1234567 microsat1 GTC G,GTCT 50 PASS NS=3;DP=9;AA=G GT:GQ:DP 0/1:35:4 0/2:17:2 1/1:40:3
```


Variant Calling Format

FILTER

```
##fileformat=VCFv4.4
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```

Sample
info

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	NA000001	NA000002	NA000003
20	14370	rs6054257	G	A	29	PASS	NS=3;DP=14;AF=0.5;DB;H2	GT:GQ:DP:HQ	0 0:48:1:51,51	1 0:48:8:51,51	1/1:43:5:.,.
20	17330	.	T	A	3	q10	NS=3;DP=11;AF=0.017	GT:GQ:DP:HQ	0 0:49:3:58,50	0 1:3:5:65,3	0/0:41:3
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20	1234567	microsat1	GTC	G,GTCT	50	PASS	NS=3;DP=9;AA=G	GT:GQ:DP	0/1:35:4	0/2:17:2	1/1:40:3

Allele Frequencies

- Allele Frequency within a sample -> reported in VCF file
 - ratio between the alleles in the observed reads
- Allele Frequency within a population -> reported in population databases (e.g.gnomAD)
 - summary statistic over multiple individuals

Genotype estimation for germline variants

- Possible genotypes for germline variants (in diploid organisms)
 - homozygot : same allele
 - heterozygot : different alleles
- Alleles encoded as **0** (= REF), **1** (=ALT1), **2** (=ALT2) in VCF files
- Phased genotypes encoded as 0|0, 0|1, 1|0, 1|1

Phasing means the genotype is ordered, so you know which is the maternal and which is the paternal allele and which alleles of different variants are being inherited together
- Unphased genotypes encoded as 0/0, 0/1 1/1

Variant Calling Models

More than just allele frequencies

- The NGS data contain errors and biases, that need to be accounted for
- Bayesian statistical approaches

$$\begin{aligned} P(G \mid D) &= \frac{P(D \mid G)P(G)}{P(D)} \\ &= \frac{P(D \mid G) P(G)}{\sum_{i=1}^n P(D \mid G_i) P(G_i)} \end{aligned}$$

where G = genotype and D = observed data (mapped reads)

Variant Calling Models

More than just allele frequencies

$$\begin{aligned} P(G \mid D) &= \frac{P(D \mid G)P(G)}{P(D)} \\ &= \frac{P(D \mid G) P(G)}{\sum_{i=1}^n P(D \mid G_i) P(G_i)} \end{aligned}$$

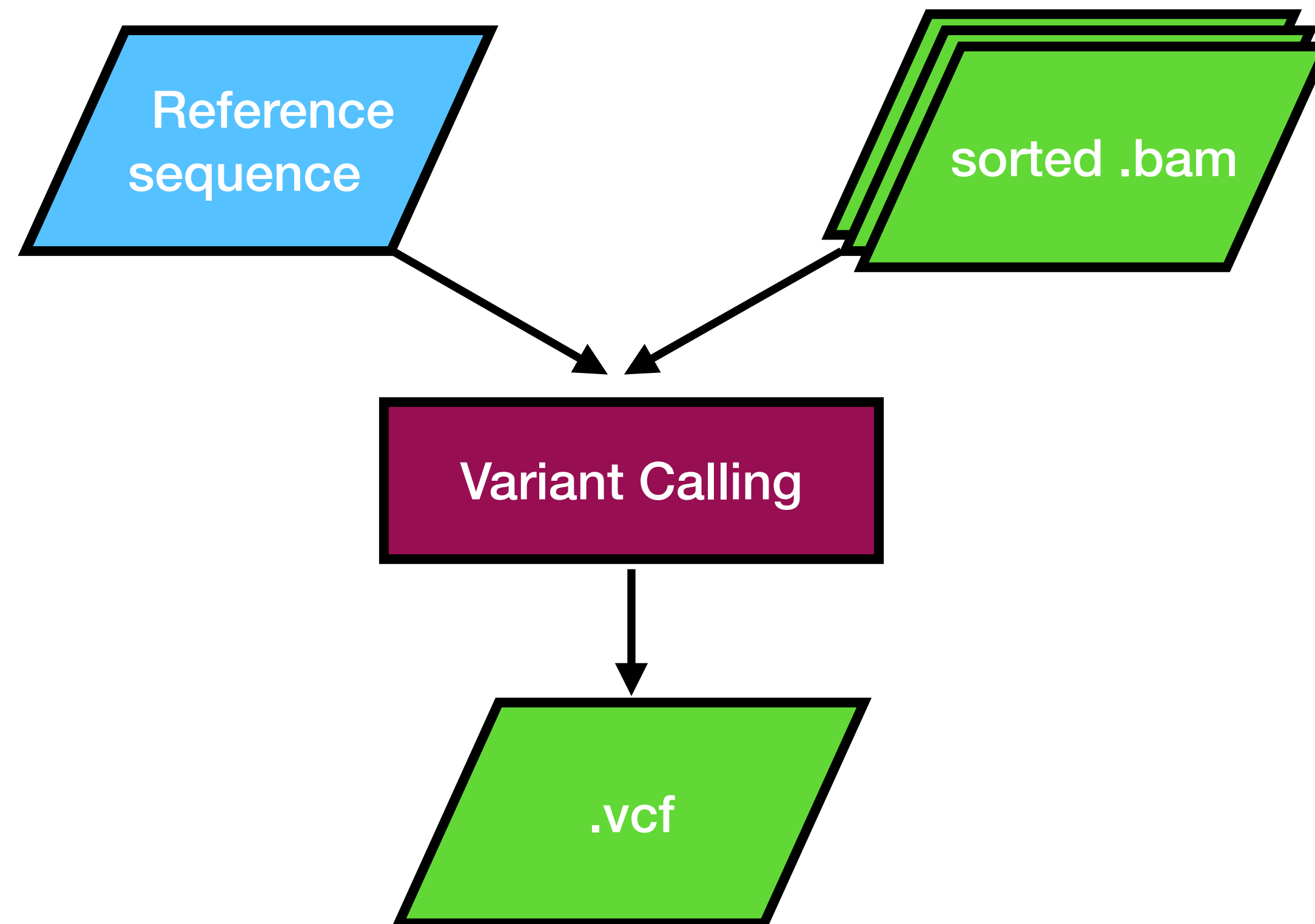
- $P(D/G)$ - error model for the data observation
 - Phred scores are integrated
 - base quality recalibration
- $P(G)$ - prior genotype probability
 - based on available genome data
 - population wide allele frequencies

Further reading: <https://arxiv.org/pdf/1207.3907.pdf>

Variant Calling Tools

- Open source
 - GATK
 - Freebayes
 - samtools
- Commercial
 - Dragen (Illumina), Sentieon, CLC Genomics (Qiagen),
 - Varvis (Rostock), Seamless (Leipzig)

Variant Calling Workflow (simplified)



Workflow (simplified)

Variant calling with samtools / bcftools

1. Index the reference genome

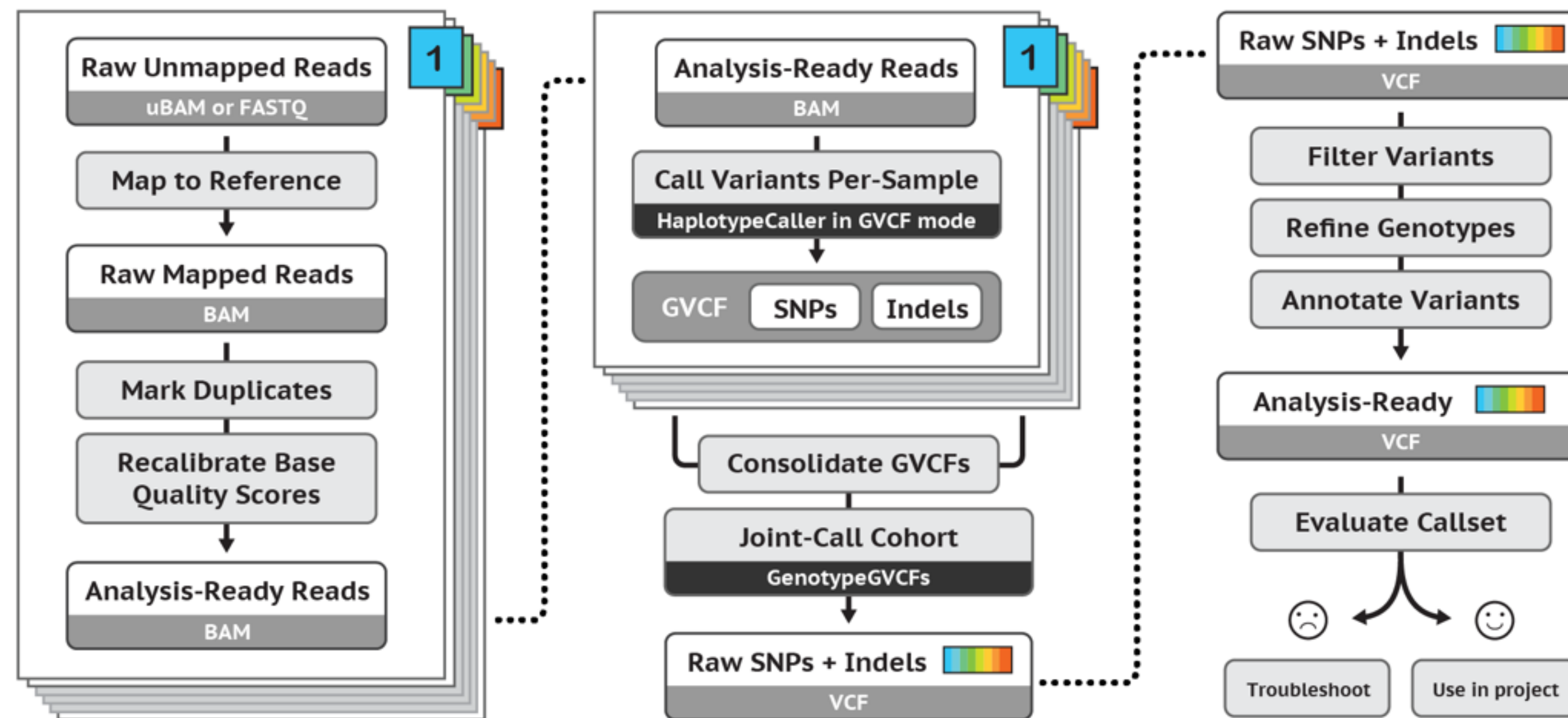
```
samtools faidx reference.fa
```

2. Run

```
bcftools mpileup -f reference.fa sample1-sort.bam  
sample2-sort.bam | bcftools call -c -v -O v -o  
variants.vcf
```

Workflow (Golden Standard)

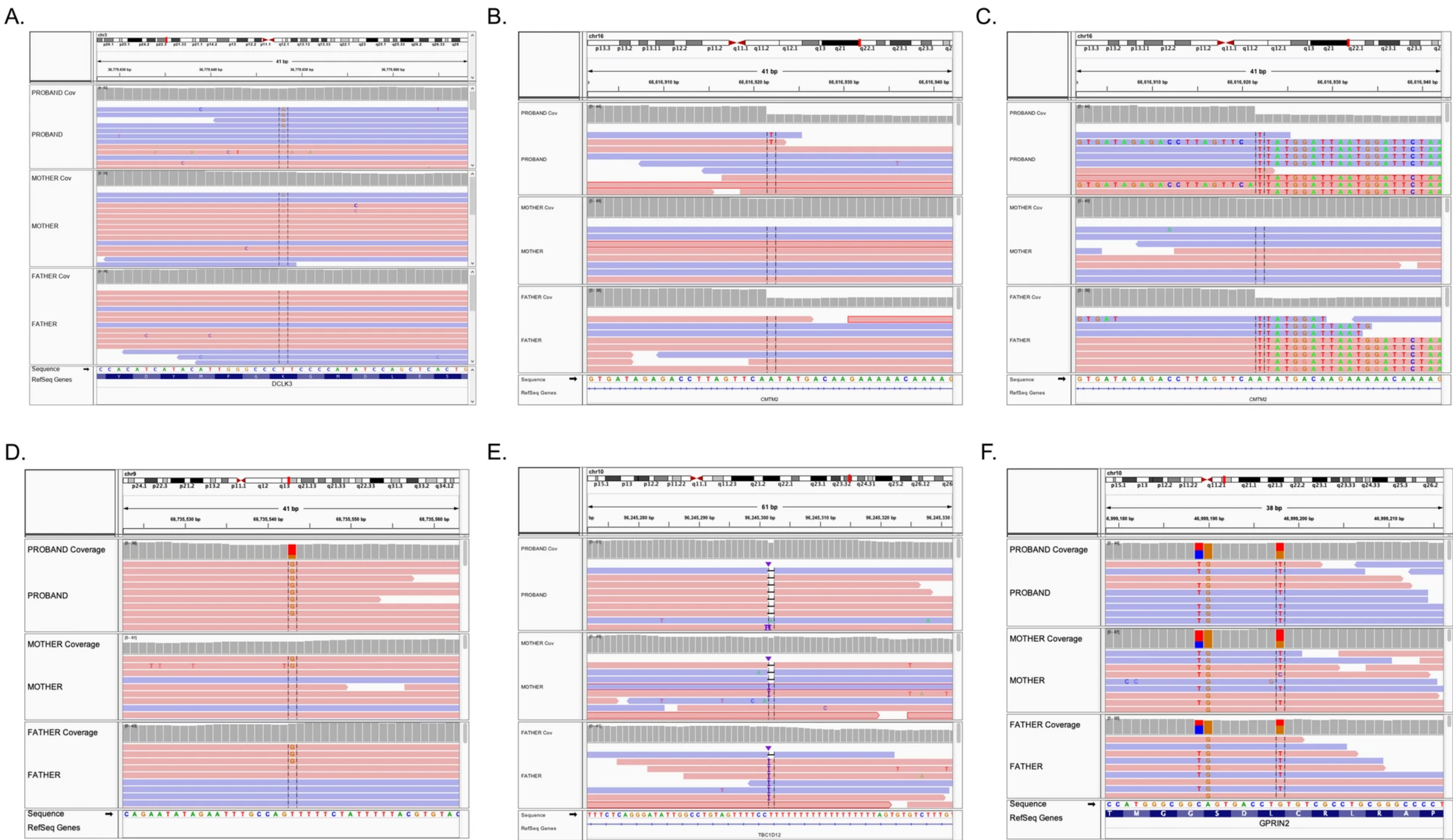
GATK Best practices



*Best Practices for SNP and Indel discovery in germline DNA
- leveraging groundbreaking methods for combined power
and scalability.*

Source: <https://gatk.broadinstitute.org>

Common artefacts in NGS alignments



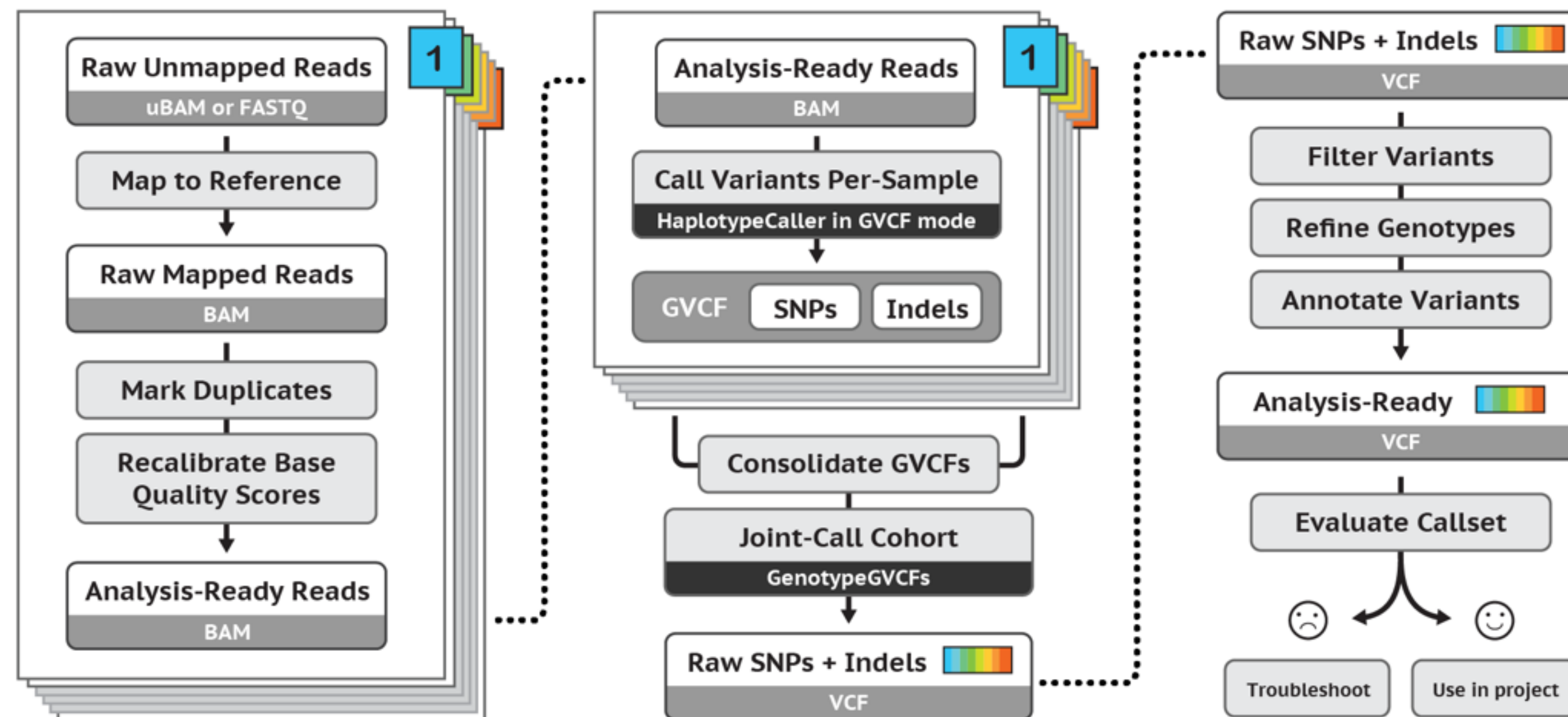
Koboldt, D.C. Best practices for variant calling in clinical sequencing. *Genome Med* **12**, 91 (2020).
<https://doi.org/10.1186/s13073-020-00791-w>

Common artefacts in NGS alignments

- A** False positive associated with low base quality. Most reads with low base quality, few reads with good quality
- B** False positive due to misalignments near the start or end of reads. Variant only covered by beginning or end of reads
- C** The same position as in **b**, but with soft-clipped bases shown in color.
- D** False positive associated with strand bias. All but one variant-supporting reads are on the reverse strand, whereas reference-supporting reads are equally represented on both strands.
- E** False positives associated with low-complexity sequences (e.g. single-base deletion at a T-homopolymer)
- F** False positives due to paralogous alignments of reads from regions not well represented in the reference.

Workflow (Golden Standard)

GATK Best practices



*Best Practices for SNP and Indel discovery in germline DNA
- leveraging groundbreaking methods for combined power
and scalability.*

Source: <https://gatk.broadinstitute.org>

Variant Annotation

- Setting the variant into a biological context
 - **position within a genomic feature** (exon/intron/promoter/intergenic)
 - **effect of the variant on the cDNA/proteins:**
 - missense (SNP leading to a non-synonymous amino acid change)
 - frameshift (InDel leading to a change in the frameshift)
 - premature stop (variant leading to a premature stop codon)
 - silent (SNP leading to no amino acid change)

Variant Annotation

- Setting the variant into a biological context
 - **position within a genomic feature** (exon/intron/promoter/intergenic)
 - **effect of the variant on the cDNA/proteins:**
 - **comparison with existing data bases**, in order to obtain allele frequency in large population
- additional input for this step is the genome annotation in gff/gtf format
- we will use SnpEff for this step

Variant Filtering

Reducing the number of variants by applying certain thresholds

- increase confidence
 - variant quality
 - read depth
 - genotype quality
- remove common variants (high frequency variants in big populations)
 - especially for human

Variant Filtering

Reducing the number of variants by applying certain thresholds

- consider predicted effect
 - remove silent/synonymous mutations
 - check pathogenicity
- filter for variants in certain regions
 - restrict to relevant genes or genomic region

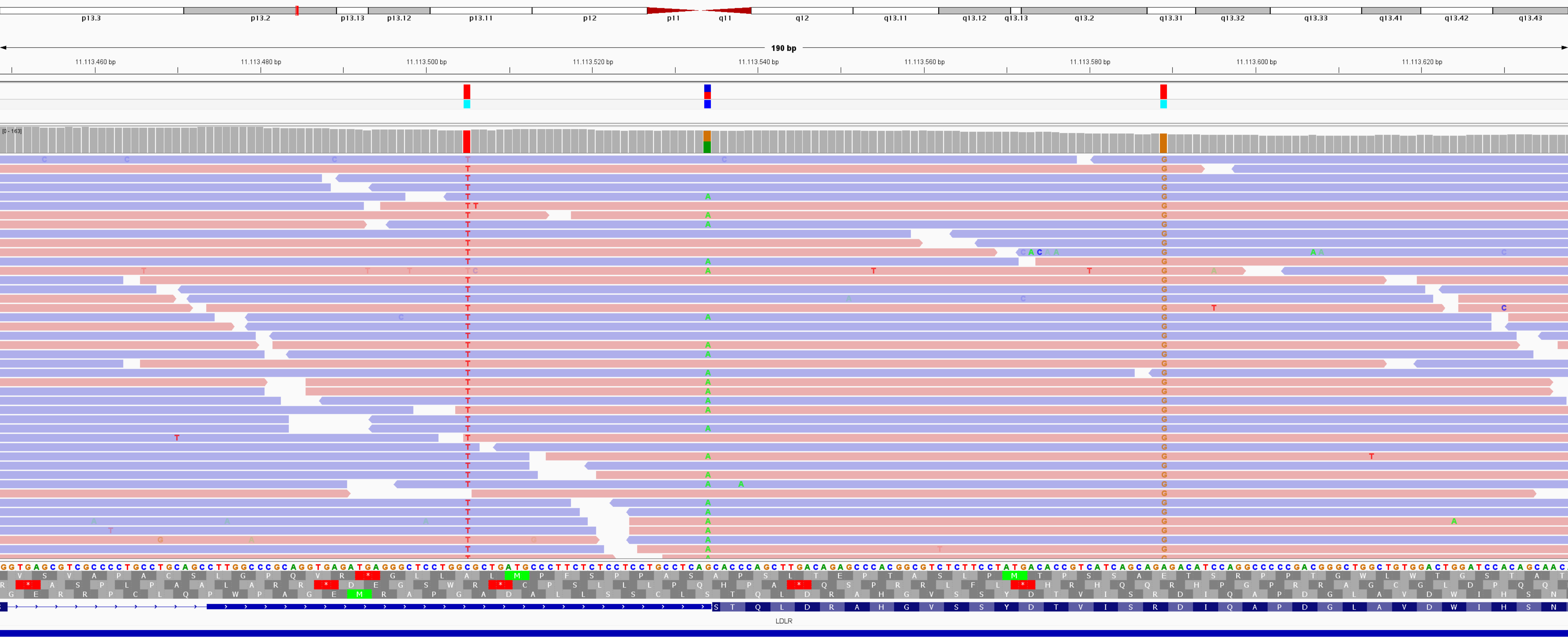
Finally: a list of candidate variants

Application of variant calling

- whole genome (re-sequencing) projects
- target enrichment projects (whole exome sequencing, custom gene panel)
- widely used in human genetics (germline variants, trio analyses) and molecular pathology/oncology (somatic variants)
- population genetics and genome wide association studies
- QTL (quantitative trait locus) identification of economically relevant traits
- evolution research: estimation of substitution rates, detection of signs of selection

Variant Calling in Diagnostics

A case of familial hypercholesterolemia



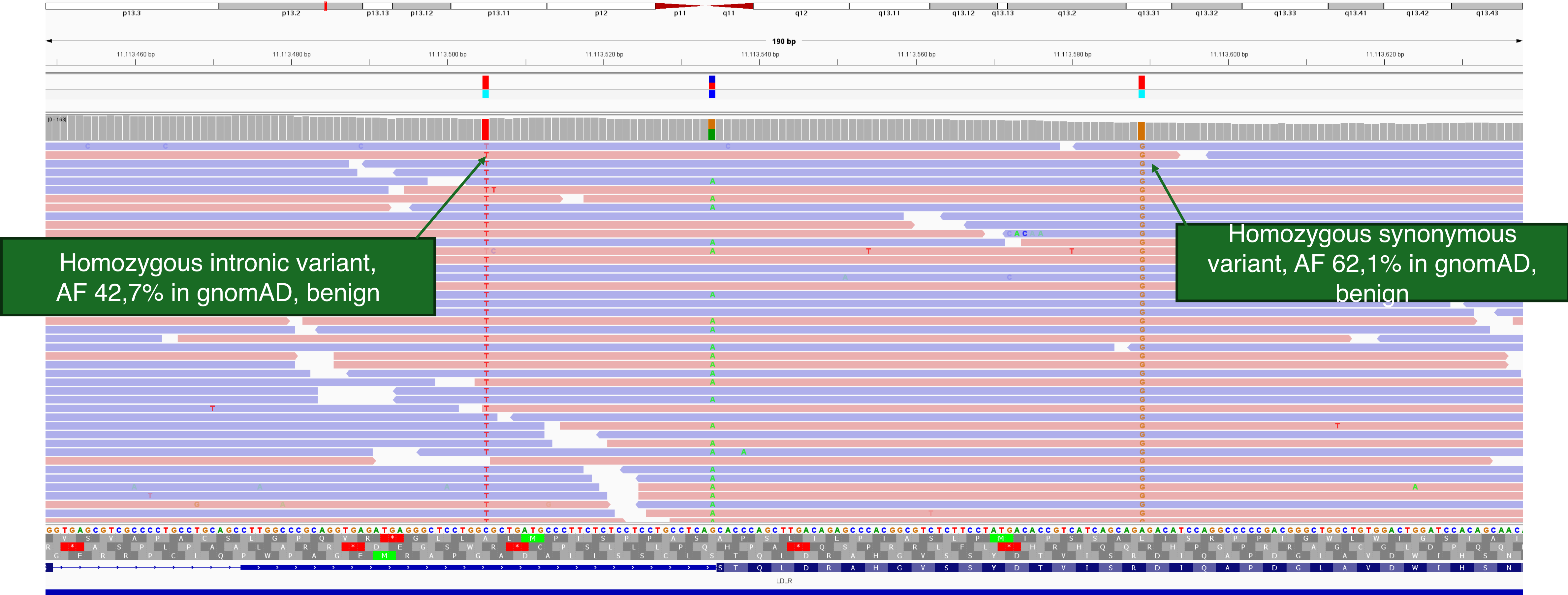
Variant Calling in Diagnostics

Germline Variants

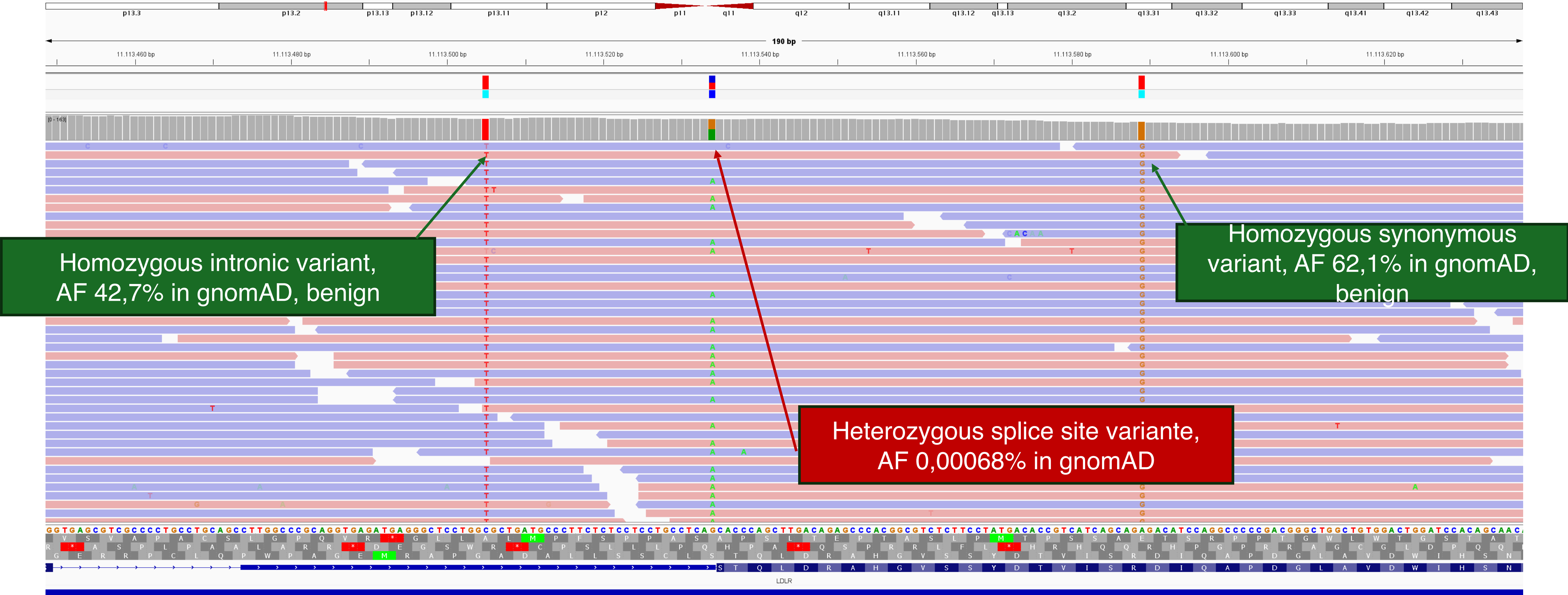
- ACMG Criteria
 - regularly updated
 - guidelines for clinical reporting

	Benign		Pathogenic			
	Strong	Supporting	Supporting	Moderate	Strong	Very Strong
Population Data	MAF is too high for disorder <i>BA1/BS1</i> OR observation in controls inconsistent with disease penetrance <i>BS2</i>			Absent in population databases <i>PM2</i>	Prevalence in affecteds statistically increased over controls <i>PS4</i>	
Computational And Predictive Data		Multiple lines of computational evidence suggest no impact on gene /gene product <i>BP4</i> Missense in gene where only truncating cause disease <i>BP1</i> Silent variant with non predicted splice impact <i>BP7</i>	Multiple lines of computational evidence support a deleterious effect on the gene /gene product <i>PP3</i>	Novel missense change at an amino acid residue where a different pathogenic missense change has been seen before <i>PM5</i> Protein length changing variant <i>PM4</i>	Same amino acid change as an established pathogenic variant <i>PS1</i>	Predicted null variant in a gene where LOF is a known mechanism of disease <i>PVS1</i>
Functional Data	Well-established functional studies show no deleterious effect <i>BS3</i>		Missense in gene with low rate of benign missense variants and path. missenses common <i>PP2</i>	Mutational hot spot or well-studied functional domain without benign variation <i>PM1</i>	Well-established functional studies show a deleterious effect <i>PS3</i>	
Segregation Data	Non-segregation with disease <i>BS4</i>		Co-segregation with disease in multiple affected family members <i>PP1</i>	Increased segregation data →		
De novo Data				<i>De novo</i> (without paternity & maternity confirmed) <i>PM6</i>	<i>De novo</i> (paternity & maternity confirmed) <i>PS2</i>	
Allelic Data		Observed in <i>trans</i> with a dominant variant <i>BP2</i> Observed in <i>cis</i> with a pathogenic variant <i>BP2</i>		For recessive disorders, detected in <i>trans</i> with a pathogenic variant <i>PM3</i>		
Other Database		Reputable source w/out shared data = benign <i>BP6</i>	Reputable source = pathogenic <i>PP5</i>			
Other Data		Found in case with an alternate cause <i>BP5</i>	Patient's phenotype or FH highly specific for gene <i>PP4</i>			

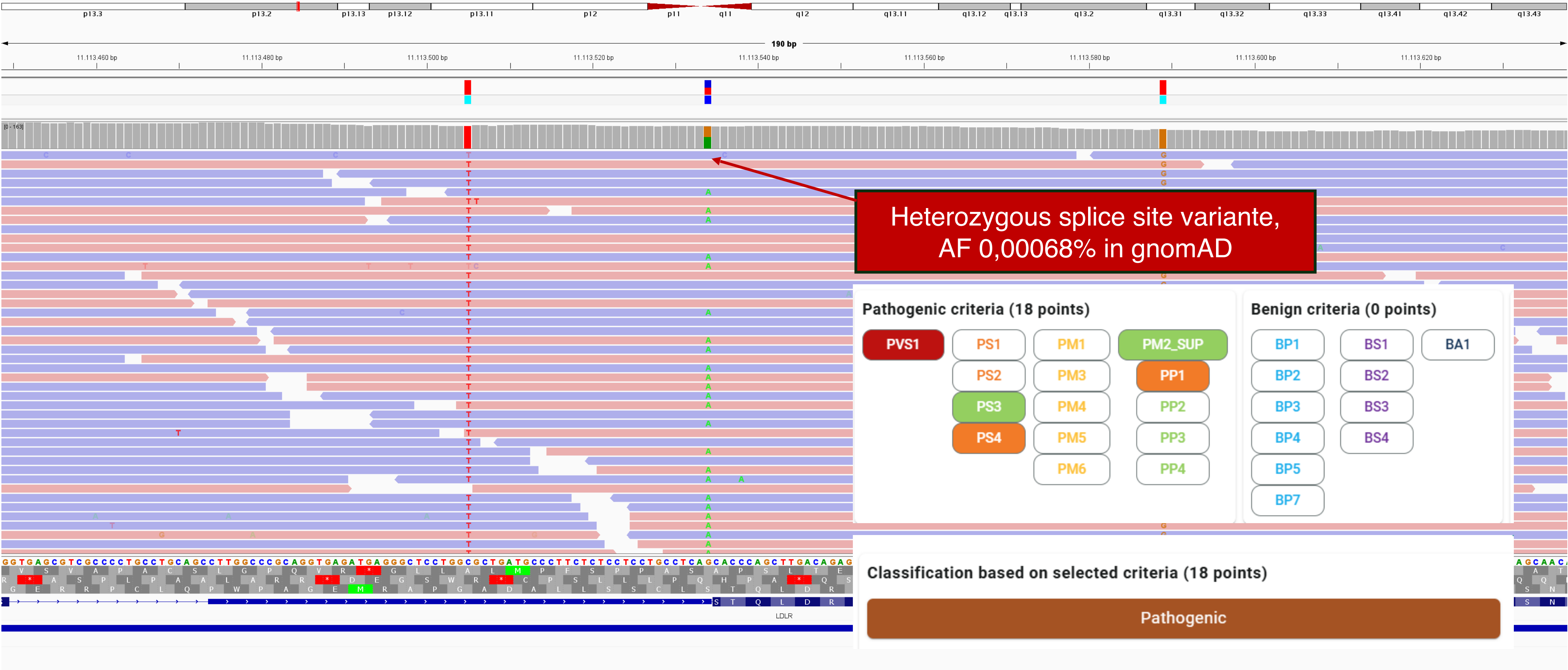
Back to the example



Back to the example

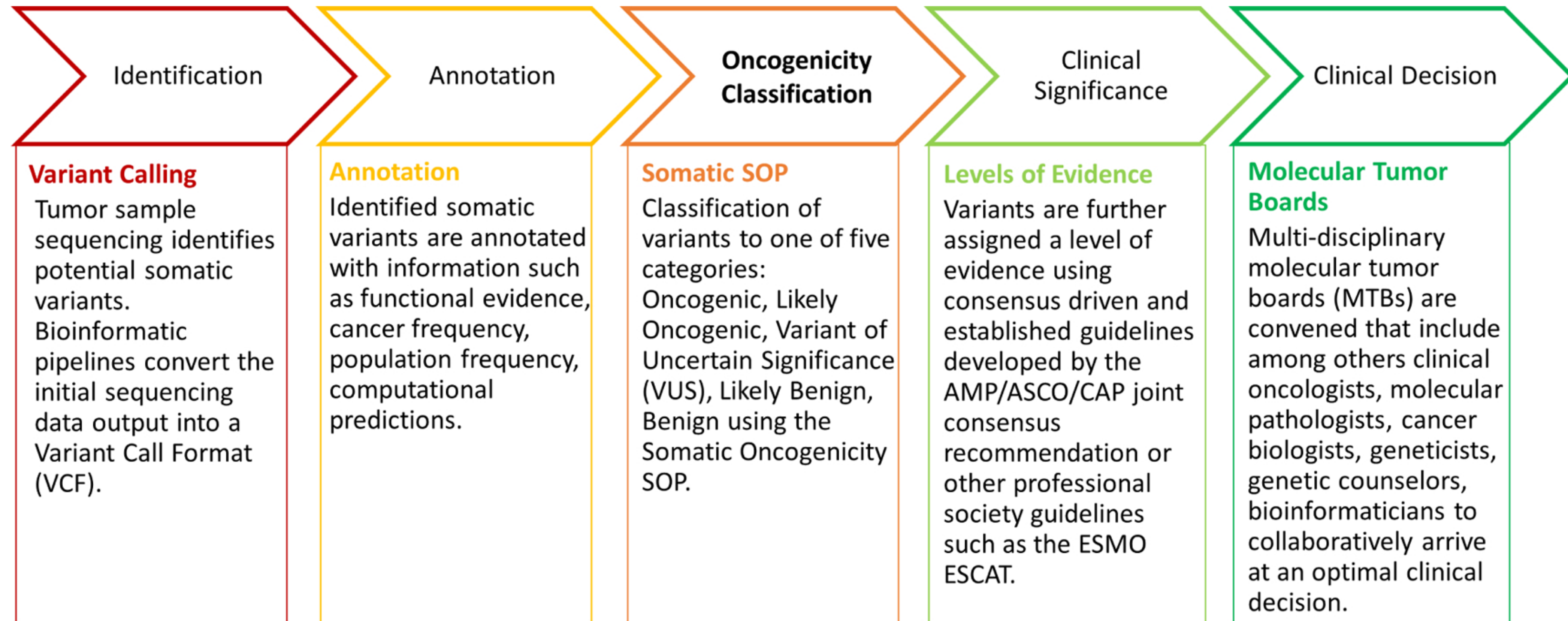


Back to the example



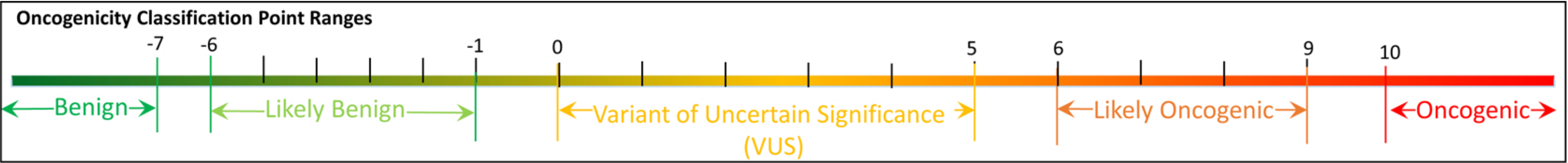
Variant Calling in Diagnostics

Somatic Variants



Variant Calling in Diagnostics

Somatic Variants



	Benign			Oncogenic			
Evidence Strength	Very Strong	Strong	Supporting	Supporting	Moderate	Strong	Very Strong
POINTS	-8	-4	-1	+1	+2	+4	+8
Population Data	MAF is >5%	MAF is >1%		Absent in population databases			
Functional Data		Well-established functional studies show no oncogenic effects				Well-established functional studies supportive of an oncogenic effect	
Predictive Data			Silent mutation (no predicted impact on splicing)		Missense change at an amino acid residue where a different missense change determined to be oncogenic has been documented	Same amino acid change as a previously established oncogenic mutation	Null variant in tumor suppressor
Cancer Hotspots				Cancer hotspots with low frequency of recurrence	Cancer hotspot with moderate frequency of recurrence	Cancer hotspot with high frequency of recurrence	
Computational Evidence			All utilized lines of computational evidence suggest no impact of a variant	All utilized lines of computational evidence support oncogenicity			

Vocabulary in Diagnostics

- **Primary analysis**
 - Demultiplexing
 - Data QC
- **Secondary analysis**
 - Mapping
 - Variant Calling
- **Tertiary analysis**
 - Variant Interpretation

Ressources for tertiary analysis

- Very often commercial solutions
- Qiagen Clinical Insights (QCI), Illumina Connected Insights, Sophia Genetics
- Varsome: <https://varsome.com/>
- Franklin: <https://franklin.genoox.com/clinical-db/home>